

Solving Matrix Games

Book:

Algorithmic Game Theory

Lemke-Howson Alg.

Finding NE (Review)

		P2	
		stag	hare
P1	stag	10, 10 ↑	0, 6 ↓
	hare	6, 0 ↑	5, 5 ↓

Nash Equilibrium

Every player plays a best response

$M \equiv$ set of actions for P1

$N \equiv$ set of actions for P2

Bimatrix Games

$$A = \begin{bmatrix} 10 & 0 \\ 6 & 5 \end{bmatrix}$$

$$B = \begin{bmatrix} 10 & 6 \\ 0 & 5 \end{bmatrix}$$

$x \equiv$ strategy for P1
 $y \equiv$ strategy for P2

payoffs: P1: $x^T A y$
P2: $x^T B y$

Calculate Mixed NE

To get another player to play a mixed NE, you must make them indifferent between their actions.

P1 indifferent

payoff for P1 playing 1 $\rightarrow$ u
payoff for P1 playing 2 $\rightarrow$ u

$$\begin{bmatrix} u \\ u \end{bmatrix} = \begin{bmatrix} 10 & 0 \\ 6 & 5 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

$$y_1 + y_2 = 1$$

$y_1 = \frac{5}{9} \quad y_2 = \frac{4}{9}$

P2 indifferent

$$\begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 10 & 6 \\ 0 & 5 \end{bmatrix} = \begin{bmatrix} v & v \end{bmatrix}$$

$$x_1 + x_2 = 1$$

$x_1 = \frac{5}{9} \quad x_2 = \frac{4}{9}$

$$M = \{1, 2, 3\}$$

$$N = \{4, 5\}$$

$$A = \begin{bmatrix} 3 & 3 \\ 2 & 5 \\ 0 & 6 \end{bmatrix}$$

$$B = \begin{bmatrix} 3 & 2 \\ 2 & 6 \\ 3 & 1 \end{bmatrix}$$

Pure NE: $x = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad y = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

Mixed NE

P1 indifferent

$$\begin{bmatrix} u \\ u \\ u \end{bmatrix} = A y$$

$$y_4 + y_5 = 1$$

4 equations 3 unknowns

In general, could have solutions, but for this A, no solutions.

No mixed strategies where P1 has all actions in the support of the NE

What if 1 of P1's actions is not in the support of the NE.

[Every action in the support of a NE must be a best response

Mathematically, in a NE:

$$x_i > 0 \Rightarrow (A\gamma)_i = u = \max \{ (A\gamma)_k : k \in M \}$$

$$y_j > 0 \Rightarrow (x^T B)_j = v = \max \{ (x^T B)_k : k \in N \}$$

$$M = \{1, 2, 3\}$$

$$N = \{4, 5\}$$

$$A = \begin{matrix} & \begin{matrix} 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} & \begin{bmatrix} 3 & 3 \\ 2 & 5 \\ 0 & 6 \end{bmatrix} \end{matrix}$$

$$B = \begin{matrix} & \begin{matrix} 4 & 5 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} & \begin{bmatrix} 3 & 2 \\ 2 & 6 \\ 3 & 1 \end{bmatrix} \end{matrix}$$

Pure NE: $x = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ $\gamma = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

Mixed NE: $x = \left[\frac{4}{5}, \frac{1}{5}, 0 \right]^T$ $\gamma = \left[\frac{2}{3}, \frac{1}{3} \right]^T$
 $A\gamma = \begin{bmatrix} \frac{3}{2} \\ \frac{3}{2} \end{bmatrix}$ $x^T B = \left[\frac{14}{5}, \frac{14}{5} \right]$

$x = \left[0, \frac{1}{3}, \frac{2}{3} \right]^T$ $\gamma = \left[\frac{1}{3}, \frac{2}{3} \right]^T$
 $A\gamma = \begin{bmatrix} 3 \\ 4 \\ 4 \end{bmatrix}$ $x^T B = \left[\frac{8}{3}, \frac{8}{3} \right]$

Hard part of finding NE in general Matrix Games: Finding Support

First, narrow scope to avoid cases with possibly infinite NE

Def. A two-player game is called nondegenerate if no mixed strategy of support size k has more than k pure best responses.

Example of degenerate

$$A = \begin{bmatrix} 3 & 3 \\ 2 & 5 \\ 0 & 6 \end{bmatrix} \begin{bmatrix} \frac{2}{3} \\ \frac{1}{3} \end{bmatrix} \quad B = \begin{bmatrix} 3 & 2 \\ 2 & 6 \\ 3 & 1 \end{bmatrix}$$

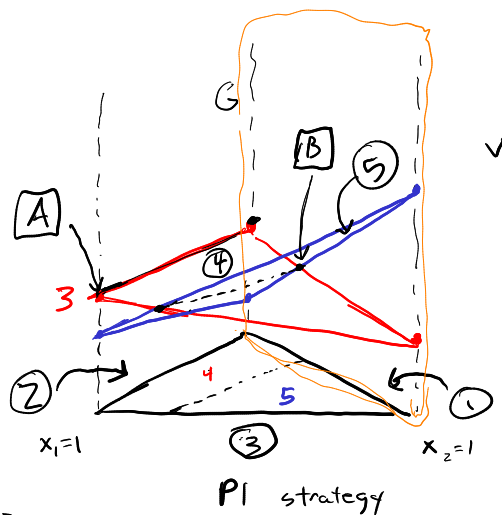
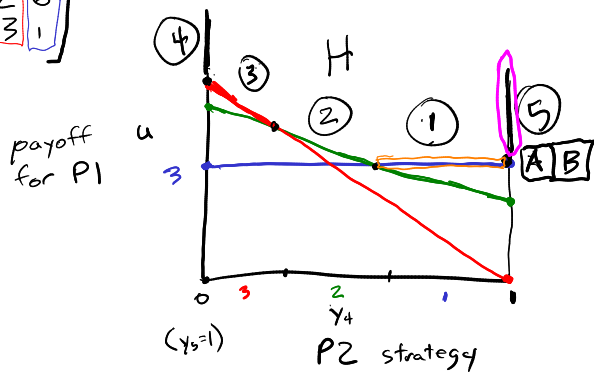
$\begin{bmatrix} 3 \\ 3 \\ 3 \end{bmatrix}$ Degenerate

$$A = \begin{bmatrix} 3 & 3 \\ 2 & 5 \\ 0 & 6 \end{bmatrix}$$

$$B = \begin{bmatrix} 3 & 2 \\ 2 & 6 \\ 3 & 1 \end{bmatrix}$$

Equilibria via Labeled Polytopes

Best-Response Diagram



$$H = \{(y_4, y_5, u) : Ay \geq u, y_4 + y_5 = 1, y_4, y_5 \geq 0\}$$

$$G = \{(x_1, x_2, x_3, v) : x^T B \geq v, \sum x_i = 1, x_i \geq 0\}$$

Each vertex gets action labels (e.g. ①)

A vertex gets label i if

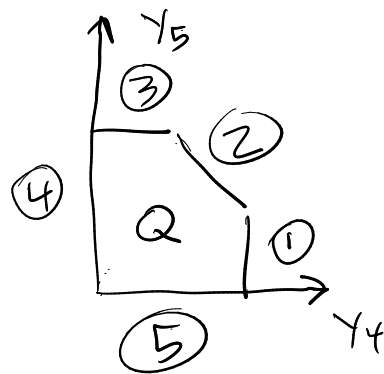
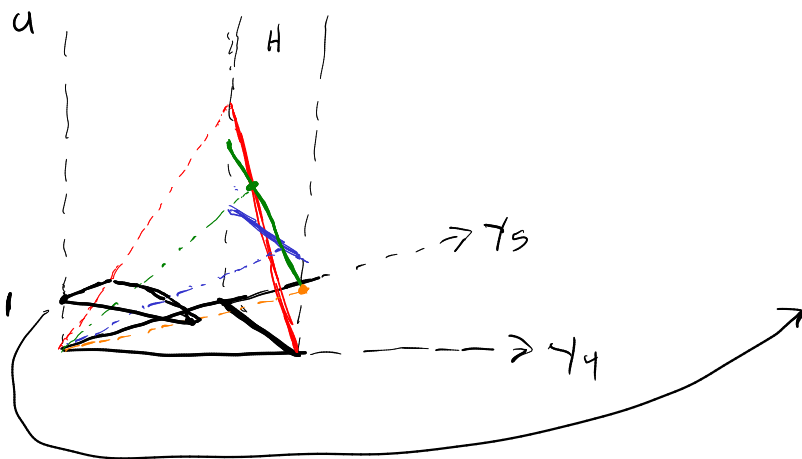
- own action i has nonzero prob
- other player's action i has zero prob

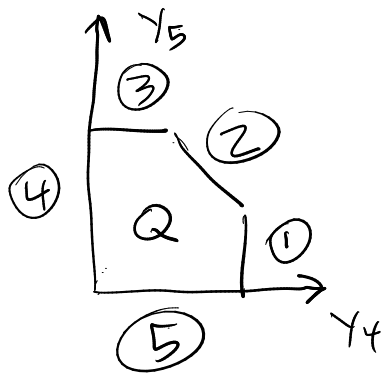
A ① ⑤ ② ③ ④ ← is a NE

B ① ⑤ ④ ③ ① ← is not a NE

A pair of vertices that is jointly fully-labelled is a NE.

Want Bounded Best Response Polytopes



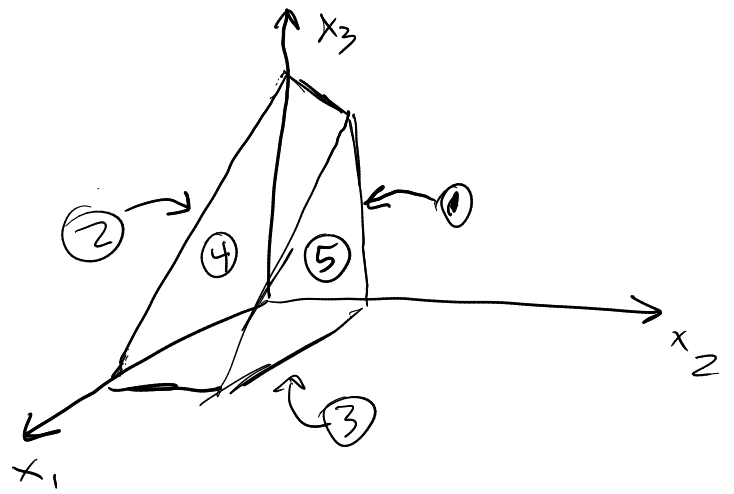


All NE

Examples:
1rs Nash

EEE

Enumeration of
Extreme Equilibria



1 NE

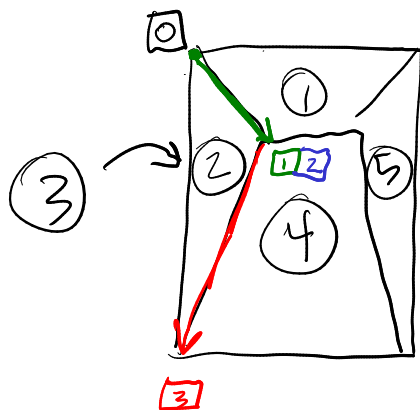
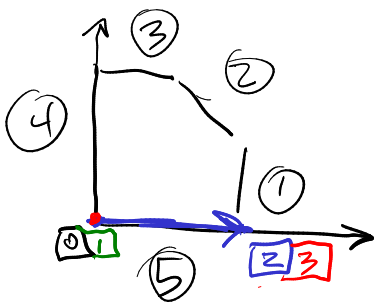
Examples:

Lemke-Howson 1964

Porter-Nudelman-Shoham
(PNS)

Algorithm 3.6 (Lemke-Howson) *Input:* Nondegenerate bimatrix game. *Output:* One Nash equilibrium of the game. *Method:* Choose $k \in M \cup N$, called the *missing label*. Let $(x, y) = (\mathbf{0}, \mathbf{0}) \in P \times Q$. Drop label k (from x in P if $k \in M$, from y in Q if $k \in N$). **Loop:** Call the new vertex pair (x, y) . Let l be the label that is picked up. If $l = k$, terminate with Nash equilibrium (x, y) (rescaled as mixed strategy pair). Otherwise, drop l in the other polytope and repeat.

Start at origin, swap labels until fully-labeled point.



0	4	5	1	2	3
1	4	5	2	4	1
2	1	5	2	4	1
3	1	5	2	4	3

NE